Lab 53 — Safety Procedures, ESD Control and Environmental Compliance

> **Course:** CompTIA Certified A+ Training (Core 1 and Core 2) (TGS-2024048317)
> **Topic 09:** Operational Procedures — Core 2, 21% of Core 2
> **Exam objective:** Apply ESD, electrical and personal safety procedures, and comply with environmental requirements for disposal and equipment protection (Core 2 objectives 4.4 and 4.5).

Goal

Safety procedures are examinable and are the part of the job where a mistake injures someone or destroys equipment. You build the complete procedure covering ESD, electrical safety, personal safety and environmental compliance, and identify hazards in a real workspace.

What you'll produce

A workspace safety procedure covering ESD, electrical, personal and environmental requirements, with a completed hazard assessment.

Tools and equipment

Anti-static equipment, safety reference material, MSDS/SDS documentation, workspace

Safety

- Disconnect mains power and hold the power button for 15 seconds before working inside any machine.
- Wear an anti-static strap connected to an earthed point; hold boards and cards by their edges.
- **Never open a power supply unit or a CRT monitor** — both retain a lethal charge after being unplugged.
- Never puncture, compress or charge a swollen lithium battery; isolate it and follow hazardous disposal procedure.

Step-by-step

1. Record what electrostatic discharge is and why it matters: a person can carry thousands of volts of static, and components are damaged at levels far below what a person can feel.
2. Record the ESD controls: an anti-static wrist strap connected to earth, an anti-static mat, anti-static bags for storage, and controlling humidity since dry air increases static build-up.

3. Record the handling rules: hold expansion cards by their edges, never touch gold contacts or chip pins, and equalise potential by touching the chassis before touching a component.
4. Record the electrical safety rules: disconnect mains power before working inside a machine, hold the power button for 15 seconds to discharge, and never open a power supply or a CRT monitor.
5. Record why a power supply and a CRT are singled out: both retain a lethal charge in their capacitors long after being unplugged.
6. Record the fire safety rules: use a Class C extinguisher on an electrical fire, never water, and know where the extinguisher and the power cut-off are before you need them.
7. Record the personal safety rules: correct lifting technique using the legs with a straight back, safety goggles when using compressed air, and an air filtration mask when cleaning dust or toner.
8. Record the MSDS or SDS requirement: every hazardous material has a safety data sheet stating its hazards, its handling requirements and its disposal method, and it must be accessible to staff.
9. Record the disposal requirements: batteries, toner cartridges, CRT monitors and electronic waste must all go to approved recycling and never into general waste.
10. Record the environmental controls: temperature and humidity management, dust control with compressed air or an anti-static vacuum, and adequate ventilation.
11. Record the power protection controls: a surge suppressor against spikes, and an uninterruptible power supply against sags, brownouts and outages, giving time for an orderly shutdown.
12. Perform a hazard assessment of your actual workspace, listing every hazard found, its risk level and the specific control you would implement.

Test it — verification

Your procedure covers ESD, electrical, personal and environmental safety; it explicitly forbids opening a PSU and a CRT and states why; Class C is named for electrical fires; and the hazard assessment lists real hazards with a control for each.

Troubleshooting this lab

Symptom	What to check
A command returns "command not found"	Re-run the `apt-get install` step at the start of the lab — the playground starts with a minimal package set.
The Killercoda terminal has reset	The playground times out when idle. Reopen it and re-run the setup commands from step 1.
A browser tool will not load	Check the URL against `labs/tools.md`. All four tools run entirely client-side and need no login.

Symptom	What to check
Output differs from the guide	Record what you actually observed — your environment differs from the reference, and explaining the difference is part of the exercise.

Review questions

1. State the exam objective this lab maps to, in your own words.
2. Which single step in this lab would you perform first on a real support call, and why?
3. What evidence would you attach to a support ticket to show this work was completed correctly?
4. Name one thing that would make this procedure fail, and how you would recognise it.

Record your evidence

Complete worksheet.md as you work through this lab and keep it — the Practical Performance assessment mirrors these tasks.
